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The Function of Regulator's Empathy and Social Distance in Adolescent Interpersonal Emotion Regulation Effectiveness: A Dyadic Approach

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Previous studies have indicated that adolescents are susceptible to emotional cues and can benefit from peers' interpersonal emotion regulation (IER). However, it remains unclear how an adolescent regulator's personal traits shape an effective IER. The present study examined the role of regulator's trait empathy and social distance between the regulator and the target on IER effectiveness among adolescents and explored the underlying behavioral mechanism. A total of 420 adolescent dyads with ages ranging from 12 to 18 years participated (212 dyads of friends; 208 dyads of strangers). After reporting their empathy levels, each regulator inferred the emotional intensity of the target based on the given negative events the target had experienced and then wrote down regulation strategies. Targets rated their own emotions before and after reading regulation strategies and evaluated the suitability of the strategies for them. The results showed that regulators with higher cognitive empathy were better at accurately perceiving targets' negative emotions, which, in turn, enhanced their regulation effectiveness. Similarly, regulators' behavioral empathy was positively related to regulation effectiveness through target-perceived strategy suitability. Additionally, close social distance enhanced the role of regulator's cognitive empathy in emotion perception accuracy and strengthened the impact of the regulator's behavioral empathy on regulation effectiveness. In contrast, closer social distance weakened the positive effect of regulator's behavioral empathy on target-perceived strategy suitability. These findings first highlight how and when different components of an adolescent regulator's trait empathy are linked to IER effectiveness, emphasizing the importance of adopting a dyadic design in the field of IER.

Keywords: interpersonal emotion regulation, social distance, trait empathy, emotion perception accuracy, adolescents

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The emotional health of adolescents has garnered great attention from psychologists. Due to the early maturation of brain regions related to emotional processing and the relatively delayed development of areas involved in cognitive regulation (Casey, 2015), adolescents experience emotions more intensely and frequently than children and adults (Larson et al., 1980), often struggling with emotional regulation (Lee et al., 2014). Emotion dysregulation during adolescence has been linked to internalizing symptoms (Pozzi et al., 2021). Given the crucial role of social interactions in both physical and mental health (Uchino et al., 2018), psychologists are increasingly focusing on interpersonal emotion regulation (IER). IER is a process in which the regulator intentionally seeks to influence or change the emotional response of a target (Gross, 2015; Nozaki & Mikolajczak, 2020; Reeck et al., 2016; Zaki & Williams, 2013). During adolescence, individuals are particularly attuned to their peers' emotions and behaviors (Andrews et al., 2021). Therefore, IER from peers can serve as a supportive source, enhancing adolescents' own emotional health (for review, see Sahi et al., 2023).

Although research on IER has increased, most studies focus on adults. Specifically, research on the factors and mechanisms influencing peer IER in adolescents, using a dyadic paradigm with high ecological validity, is scarce. This study addresses this gap by examining how the regulator's trait empathy and the perceived social distance between the regulator and target influence IER effectiveness and related behavioral mechanisms.

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Interpersonal Emotion Regulation and Related Theoretical Model

In contrast to intrapersonal emotion regulation, where the regulator and target are the same individual, IER involves two distinct individuals, with the regulator's actions influencing the target's emotional responses (Gross, 2015; Zaki & Williams, 2013). As the initiator of the regulation, the regulator plays a central role in this process. The Extended Process Model of Extrinsic Emotion Regulation, a widely used theoretical framework, describes the process by which the regulator regulates the target's emotions in social interactions (Nozaki & Mikolajczak, 2020; Reeck et al., 2016). This model consists of three interconnected stages.

The first is the identification stage, where the regulator recognizes the target's emotions in a situation and sets a regulation goal. A potential failure point in this stage is inaccurate emotion perception, which can hinder effective regulation (Nozaki & Mikolajczak, 2020). The second stage, also known as the strategy formulation stage, involves the regulator considering potential strategies to influence the target's emotions and selecting the most appropriate one based on the target's needs. The final stage is the implementation stage, where the regulator enacts the chosen strategy. A key factor in the second and third stages is the target's perception of the strategy's suitability (Dixon-Gordon et al., 2015; Nozaki & Mikolajczak, 2020), as the success of the IER largely depends on the target's acceptance of the strategy.

Regulator's Empathy and Interpersonal Emotion Regulation Effectiveness

Among the various factors that contribute to the effectiveness of IER, the regulator's empathy plays an indispensable role. Empathy is a complex construct that can be viewed from the perspectives of trait empathy, state empathy, and empathic ability (Mayukha et al., 2020). Of these, trait empathy is considered to offer a more consistent and stable account of individual differences in empathic tendencies across a wide range of contexts (Cuff et al., 2016). Therefore, the present study specifically focuses on trait empathy, which reflects enduring individual dispositions to experience and express empathy, rather than temporary emotional states or perceptual capacities (Cuff et al., 2016; Mayukha et al., 2020). For clarity and to avoid redundancy, the term "empathy" hereafter refers specifically to trait empathy. Drawing upon previous research, empathy, a critical aspect of social cognition, is defined as an individual's enduring tendency to understand and share others' emotional experiences, as well as to express the understanding and caring to them (de Waal, 2008; Lam et al., 2011; Wang et al., 2021). Some researchers have proposed from a theoretical perspective that regulator's empathy is critical for accurately perceiving the target's emotions and properly implementing strategies and can be an important antecedent for IER effectiveness (for reviews, see Reeck et al., 2016; Zaki, 2020). Supporting this, a cross-sectional study found a significant positive relationship between the regulator's empathy and the IER tendency (Chavira Trujillo et al., 2022). Moreover, an ecological momentary assessment study showed that empathy enhances goal achievement in everyday IER (Geiger et al., 2024). Additionally, neuroscientific research also supports this link, showing greater activation in empathy-related brain regions when regulators work to improve others' emotions (Hallam et al., 2014; Morawetz et al., 2021).

Although prior theoretical and empirical studies have consistently highlighted the role of the regulator's empathy in the IER effectiveness, it is important to interpret these findings cautiously, as empathy is an umbrella term that encompasses multiple components (de Waal, 2008; Lam et al., 2011; Wang et al., 2021). Empathy is generally understood to consist of affective empathy (emotional contagion, involving bottom-up process to vicariously share other's emotional experiences) and cognitive empathy (top-down process or tendency to understand others' emotions; de Waal, 2008; Lam et al., 2011; Wang et al., 2021). Some scholars have also emphasized that empathy involves not only internal experiences but also external manifestations that demonstrate understanding and concern for others' emotions. This component has been conceptualized as behavioral empathy, referring to the outward expressions of empathic motivation through verbal and nonverbal behaviors (Lam et al., 2011; Wang et al., 2021). Specifically, Lam et al. (2011) operationalized this behavioral component as verbal expressions (e.g., clearly stating understanding of another's feelings) and nonverbal actions (e.g., making gestures or taking actions to support others in distress) that signal emotional resonance with others. Unlike cognitive and affective empathy, which primarily reflect internal states and may not be easily observed by others, behavioral empathy serves as an explicit and observable indicator of trait empathy. This is particularly significant in collectivist cultures, where alignment between internal values and outward behaviors is highly valued. In IER contexts, individuals higher in behavioral empathy may be more motivated to engage in concrete, supportive behaviors to alleviate others' negative emotions.

While cognitive, behavioral and affective components are all encompassed within the umbrella term "empathy," they may play distinct roles in the regulation of others' emotions. Direct evidence linking behavioral empathy to IER outcomes remains limited. However, related research on empathic concern provides indirect support for our hypothesis. Previous empirical study has found that empathic concern positively correlates with a greater tendency to improve others' negative emotions (Chavira Trujillo et al., 2022). Although empathic concern is traditionally classified as other-oriented affective empathy (Chavira Trujillo et al., 2022), it reflects the motivation to care for others' well-being, which can often manifest behaviorally as prosocial actions (Batson, 1991; Batson et al., 1987; de Waal, 2008). Therefore, findings on the role of empathic concern in IER provide preliminary support for our expectation that individuals with higher levels of behavioral empathy may be more likely to enhance others' emotional states. However, it is important to clarify that we do not equate behavioral empathy with empathic concern. Rather, we argue that empathic concern often serves as a motivational precursor to helping or supporting behavior and thereby has relevance to the behavioral manifestation of empathy (Batson, 1991).

As for cognitive and affective empathy, emerging evidence suggests that they play distinct roles in shaping the effectiveness of IER. Studies have consistently shown that cognitive empathy is positively linked to both the goal and tendency to reduce others' negative emotions (Chavira Trujillo et al., 2022; Geiger et al., 2024; Levy-Gigi & Shamay-Tsoory, 2017). In contrast, individuals high in affective empathy may internalize the target's negative emotions, potentially leading to emotional distress and avoidance behaviors (Weisz & Cikara, 2021). Dyadic studies have shown that higher levels of affective empathy are associated with greater negative affect and

increased sympathetic nervous system reactivity in response to others' distress (Brown et al., 2020), which is linked to avoidance motivation (Sege et al., 2017). Correspondingly, previous research has observed a negative relationship between affective empathy and the tendency to improve others' negative emotions (Chavira Trujillo et al., 2022).

Due to the varying developmental levels of the three components of trait empathy in adolescence, their differential impacts on IER effectiveness may be particularly salient during this stage. Affective empathy is particularly prominent in adolescence (Wang et al., 2021), making adolescents more sensitive to peers' negative emotions (Andrews et al., 2021; Guyer et al., 2016). However, heightened emotional resonance with distressed peers may increase emotional costs, potentially reducing regulation effectiveness. In contrast, cognitive and behavioral empathy are still maturing (Wang et al., 2021), and individual differences in these developing components may influence the effectiveness of regulating other's emotions. Adolescents with higher levels of cognitive and behavioral empathy tend to focus more on others' needs rather than internalizing distress, which may enhance regulation effectiveness (Zaki, 2020).

Despite growing interest in the role of trait empathy in IER, most existing studies have focused exclusively on cognitive and affective empathy in adults, relying primarily on self-reported measures of regulation effectiveness. Few studies have directly examined behavioral empathy or investigated how different components of regulator's trait empathy influence objective regulation outcomes for targets in adolescence. The present study addresses this gap by employing a dyadic design to examine the behavioral mechanisms through which trait empathy contributes to IER effectiveness during adolescence.

The Mediating Role of Emotion Perception Accuracy Between Regulator's Cognitive Empathy and Regulation Effectiveness

According to the extended process model of extrinsic emotion regulation, the effectiveness of IER depends on the regulator's accurate perception of the target's emotions, typically measured by the absolute difference between the regulator's inferences and the target's self-reported ratings (Israelashvili et al., 2020; Zhou et al., 2017). Previous study in adults has found that regulator's emotion recognition accuracy helps provide support that matches the target's emotional needs (Pauw et al., 2025). Similarly, adolescents with lower emotion perception accuracy often struggle to maintain peer relationships (Gleason et al., 2009), likely because they are unable to respond appropriately to their peers' emotional needs. Furthermore, studies on adolescent romantic relationships have also found that emotion perception accuracy is positively correlated with relationship satisfaction as reported by the romantic partners (Berlamont et al., 2023; Haugen et al., 2008).

Reeck et al. (2016) first introduced the process model of extrinsic emotion regulation, arguing that regulators' emotion perception accuracy largely depends on their social cognition traits. Cognitive empathy refers to a relatively stable, top-down process or tendency to understand others' emotions (de Waal, 2008; Zaki, 2020), which may play a key role in emotion perception. Although both cognitive empathy and emotion perception accuracy play important roles in social interactions, they are conceptually distinct constructs. Cognitive empathy reflects a general dispositional trait, whereas emotion perception accuracy is an interpersonal outcome that largely depends on context factors, such as target's familiarity, facial expressions, tone of voice, or specific events that elicited the emotion (Zaki & Ochsner, 2011; Zaki et al., 2009). A recent meta-analysis has found a stable, significant positive relationship between cognitive empathy and emotion recognition accuracy (Qiao et al., 2025). During adolescence, emotional cues from peers can disrupt emotion recognition accuracy through emotional contagion (Kral et al., 2017), making top-down processing of peers' emotions essential for accurate perception. Empirical studies have supported that cognitive empathy is positively associated with emotion recognition accuracy in adolescents (Bek et al., 2022; Lui et al., 2016). Building on evidence that emotion perception accuracy contributes to regulation effectiveness and that cognitive empathy facilitates accurate emotion perception, we propose the following hypothesis:

> *Hypothesis 1:* Emotion perception accuracy would mediate the relationship between regulator's cognitive empathy and regulation effectiveness.

The Mediating Role of Target-Perceived Strategy Suitability Between Regulator's Behavioral Empathy and Regulation Effectiveness

According to the extended process model of extrinsic emotion regulation, the effectiveness of IER also depends on whether regulators provide appropriate strategies for their targets (Nozaki & Mikolajczak, 2020; Reeck et al., 2016). An empirical sampling study has found that proper strategy provided by peers can reduce adolescents' experience of negative emotions (Do et al., 2025). In line with this, previous studies have also demonstrated both immediate and long-term buffering effects of peer social support on psychological distress (Lyell et al., 2020). Considering that a peer's provision of a suitable regulation strategy is one of the necessary facets of social support, it can be inferred that regulation strategy suitability is a positive predictor of IER effectiveness.

To provide appropriate IER strategies, regulators must engage their behavioral empathy to accurately recognize the target's situation and needs. Although empirical research directly examining the role of behavioral empathy in effective strategies providing is limited, studies on empathic concern, which reflects a motivational tendency to care for others and may precede observable empathic behavior, offer some indirect support. For example, individuals with higher empathic concern tend to employ more adaptive regulation strategies (Chavira Trujillo et al., 2022), and adolescents reporting greater empathic concern often respond more appropriately to others' distress (Profe et al., 2021; Van der Graaff et al., 2018). Admittedly, these findings provide preliminary support for the potential role of behavioral empathy in providing effective IER strategies. However, they should be interpreted with caution, as empathic concern is conceptually distinct from behavioral empathy, which we define as the outward expression of empathic motivation through verbal and nonverbal behaviors. Thus, direct research is needed to clarify how behavioral empathy contributes to IER effectiveness. Considering the role of

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strategy suitability in IER effectiveness, we propose the following hypothesis:

Hypothesis 2: Target-perceived strategy suitability would mediate the relationship between regulator's behavioral empathy and regulation effectiveness.

The Moderating Role of Social Distance Between Regulator's Empathy and Regulation Effectiveness

Social distance is one of the important facets of psychological distance that was characterized as the perceived distance between individuals in terms of social relationships, emotional connections, and so forth (Magee & Smith, 2013). It encompasses aspects such as intimacy and the extent to which people feel connected or distanced from one another (Liberman & Trope, 1998; Soderberg et al., 2015). In experimental paradigms, close social distance is often represented by interactions between participants and their friends, while far social distance is typically defined by interactions between participants and strangers (Soderberg et al., 2015). Given that adolescents are more likely to assess the investment and return in social decision making (Padilla-Walker & Fraser, 2014) and are particularly sensitive to peer-related situational cues (Andrews et al., 2021), social distance between peers may serve as a moderator in interpersonal interactions. Moreover, adolescent cognitive and behavioral empathy are still developing and not yet fully mature (Wang et al., 2021). These empathic traits may rely on situational cues, such as closer social distance, to effectively guide their expression in social contexts. We hypothesize that social distance moderates the relationship between the regulator's empathy (specifically, cognitive and behavioral empathy) and IER effectiveness, as well as the relationship between the regulator's empathy and mediating factors.

Specifically, for adolescent dyads with closer social distance, the regulator's cognitive and behavioral empathy may more strongly enhance the target's perception of regulation effectiveness. This may be because, with closer social distance, targets are more familiar with the regulator's empathic traits, making them better able to perceive the regulator's emotional understanding and caring intentions. With regard to the potential mediating role of emotion perception accuracy, social distance may act as a moderator, such that closer social distance strengthens the impact of the regulator's cognitive empathy on emotion perception accuracy. Notably, empathy or emotion recognition requires cognitive resources (Cameron et al., 2019). According to social baseline theory, individuals engaged in emotional exchanges tend to conserve energy (Beckes & Sbarra, 2022), which may lead regulators to prefer managing the emotions of familiar targets, as doing so reduces cognitive effort and increases their sense of efficacy in emotional processing (Beckes & Coan, 2011; Beckes & Sbarra, 2022). Empirical findings support this perspective, showing that regulators exhibit greater intentionality, care, and self-efficacy when supporting emotionally close others (Tanna & MacCann, 2023). Additionally, adolescents are more responsive to the needs of socially closer peers (van de Groep et al., 2020), reinforcing the notion that closer social distance amplifies the positive effects of the regulator's empathy. Similarly, we hypothesize that social distance may moderate the mediation pathway from the regulator's behavioral empathy to the target's perception of strategy suitability. Specifically, when empathizing with a socially close target, the regulator's empathic motivation is likely to be more strongly activated, resulting in the selection of more appropriate strategies for regulating the target's emotions. Based on this reasoning, we propose the following hypotheses:

Hypothesis 3: Social distance will moderate the effects of regulator's cognitive empathy on emotion perception accuracy and regulation effectiveness. Specifically, these effects will be much stronger when social distance is close (Figure 1).

Hypothesis 4: Social distance will moderate the effects of regulator's behavioral empathy on target-perceived strategy suitability and regulation effectiveness. Specifically, these effects will be much stronger when social distance is close (Figure 2).

The Present Research

The present study examined how and when different components of the regulator's trait empathy are associated with regulation effectiveness in peer's IER. Specifically, the study tested the following: the mediating effect of emotion perception accuracy on the relationship between the regulator's cognitive empathy and regulation effectiveness (Hypothesis 1), the mediating effect of target-perceived strategy suitability on the relationship between the regulator's behavioral empathy and regulation effectiveness (Hypothesis 2), and the moderating effect of social distance on these direct and indirect relationships (Hypotheses 3 and 4). To enhance ecological validity, we

Figure 1
Proposed Moderated Mediation Model Between Regulator's Cognitive Empathy and Regulation Effectiveness

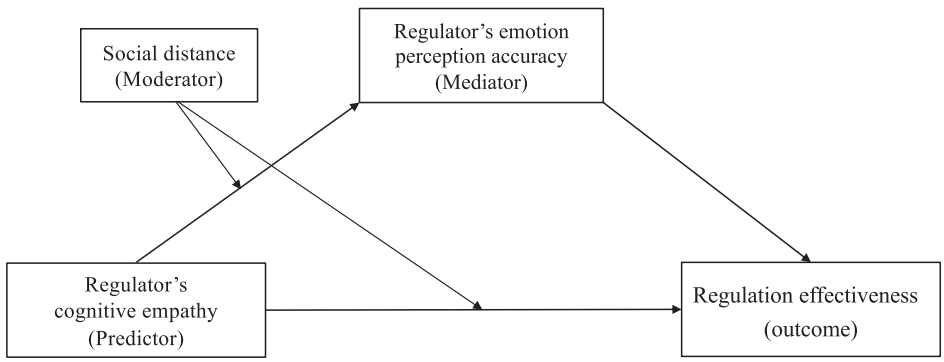

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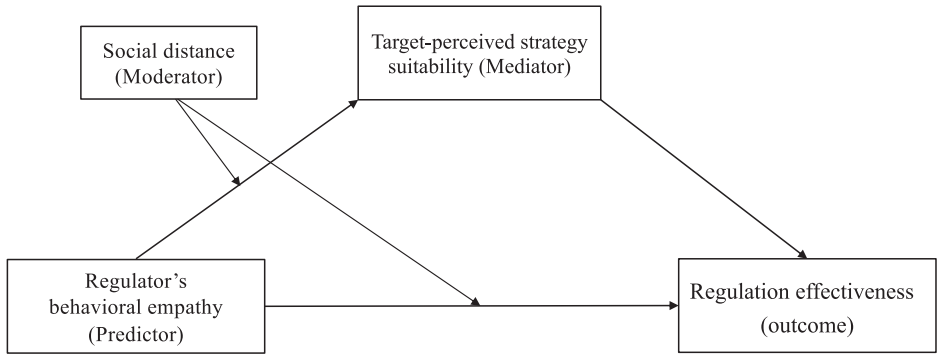
Figure 2
Proposed Moderated Mediation Model Between Regulator's Behavioral Empathy and Regulation Effectiveness



adopted a dyadic design involving both friend and stranger dyads of adolescents. Negative events commonly experienced by adolescents were used to trigger the targets' negative emotions.

Method

Participants

The final sample included 420 dyads of adolescent participants ($n_{\text{totality}} = 840$, $n_{\text{dyads}} = 420$, 66% female). Adolescent ages ranged from 12 to 18 years old. Adolescents were recruited from four middle schools via online posters. Among them, 212 dyads were same-gender friends nominated by each other from the same class ($n_{\text{friend dyads}} = 212$, 51% female, $M_{\text{age}} = 15.79$, $SD = 1.64$). The other 208 dyad participants were same-gender strangers from the same grade ($n_{\text{stranger dyads}} = 208$, 49% female, $M_{\text{age}} = 15.81$, $SD = 1.66$). Four more dyads of adolescent participants were enlisted but were not included in the analysis. They failed to follow the instructions and wanted to quit during the experiment. None of the adolescent participants had a diagnosed cognitive or sensory disorder. Additionally, the educational levels of the adolescent participants' mothers were documented and categorized into three groups: 23.5% had completed junior high school or below, 22.4% had a high school diploma, and 54.1% had obtained a bachelor's degree or higher. The study was approved by the ethics committee of the School of Psychological and Cognitive Sciences, Peking University. All the adolescents signed informed consent forms prior to the study.

Sample Size

Prior to conducting the experiment, we referred to a previous study, which suggested that a sample size of approximately 170 dyads of participants would be sufficient (Gregory et al., 2020). Given that this study includes two groups—friend dyads and stranger dyads—we opted to recruit more than twice the suggested sample size to ensure robustness.

Additionally, Monte Carlo simulation-based power analysis (Schoemann et al., 2017) was conducted post hoc to evaluate whether the sample size was sufficient for detecting effects in the mediation model. The analysis focused on the smallest observed effect size (indirect effect = 0.05) within the mediation models of the present study to ensure adequate power for capturing all effects. Specifically, we performed 2,000 simulations using a sample size of 420. The simulated model was constructed using parameter estimates derived from the results of the present study: The effect of the independent variable (X) on the mediator (M) was $a = .18$, the effect of the mediator (M) on the dependent variable (Y) was $b = 0.27$, and the direct effect of X on Y was $c' = 0.12$. To ensure replicability, a random seed of 1,234 was used. The results demonstrated that a sample size of 420 provided sufficient statistical power, achieving a power level of 0.95.

Design and Materials

Regulator's Empathy

As mentioned in the introduction, this study focuses on the impact of regulator's trait empathy on the IER effectiveness. Empathy scales are among the most widely used methods for assessing trait empathy (Luberto et al., 2018; Yavuz et al., 2024). In the present study, we adopted the Chinese version of the Griffith Empathy Measure (GEM, Q. Zhang et al., 2014) to evaluate regulators' trait empathy. The GEM was originally designed by Dadds et al. (2008) for parents to make ratings of their children's trait empathy. Each item is phrased concisely and simply, facilitating understanding for reporters. Q. Zhang et al. (2014) revised the GEM into a Chinese version and validated it within the Chinese context. Recognizing that self-report is one of the most common forms for measuring trait empathy, all items in the Chinese revision form of the GEM were modified to the first person, allowing adolescents to report on their own trait empathy.

The reliability and validity of this scale were found to be satisfactory in a sample of Chinese adolescents (Q. Zhang et al., 2014). Confirmatory factor analysis indicated acceptable model fit for both the two-factor and three-factor models, e.g., goodness of fit index ≥ 0.90, root-mean-square error of approximation ≤ 0.08, with the three-factor model demonstrating slightly superior performance on certain fit indices, such as lower root-mean-square residual and higher comparative fit index. While prior research has largely concentrated on the cognitive and affective components of empathy in regulators, the behavioral component, an explicit manifestation of the empathy trait, has been overlooked, despite its potential critical role in IER effectiveness. Therefore, in addition to examining the effect of

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cognitive and affective empathy, this study also aimed to explore the contribution of behavioral empathy to regulation effectiveness. Consequently, the three-factor model of the GEM was adopted.

The Chinese version of this scale included 17 items and comprises three subscales: The cognitive empathy subscale includes five items (e.g., "I rarely understand why other people cry" [a reverse scoring item], Cronbach's $\alpha = .73$), the behavioral empathy subscale includes six items (e.g., "I feel upset for people who are physically disabled," Cronbach's $\alpha = .76$), and the affective empathy subscale includes six items (e.g., "I get upset when seeing another person cry," Cronbach's $\alpha = .92$). The items are scored on a 9-point Likert scale, from −4 (*strongly disagree*) to +4 (*strongly agree*).

Interpersonal Emotion Regulation Task

The Interpersonal Emotion Regulation Task was adapted from previous research focused on IER in romantic relationships (Gregory et al., 2020). Additionally, a recent study employed a similar design (Arican-Dinc & Gable, 2025). Each adolescent in the pair was randomly assigned the role of either regulator or target. First, friend dyads exchanged greetings, while stranger dyads introduced themselves in turn. After 5 min, the experimenter separated them into different rooms. Next, regulators were presented with four scenarios describing negative events experienced by their targets. As the scenarios in the original paradigm were tailored for close relationships among adults and deemed unsuitable for adolescents, we revisited and recoded the scenarios to ensure their appropriateness. The four selected scenarios, as identified and coded in our previous unpublished research (Wang, 2023), are the most commonly encountered among adolescents, including "fail an important exam," "have difficulties in communicating with parents, and often be blamed by parents," "find it hard to fit in group activities, feeling excluded," and "the academic demands are overwhelming, making it difficult to find time to relax." To enhance immersion, the scenarios were narrated from the first-person perspective of the regulated individual. An example scenario was as follows:

> For example, in a recent midterm exam, my performance dropped significantly compared to before. This has left me feeling disheartened and pessimistic, making it hard to calm down and focus on addressing the gaps in my knowledge after the exam.

The regulators were asked to read each scenario and infer how emotionally negative their target would feel when encountering these events under the instruction of "One day, your friend/partner shared some of the recent negative experiences they were going through. Please rate how negativity that your partner might have felt during the moment they were experiencing the events they are describing." Meanwhile, targets read the same scenarios, written in the first person, under the instruction of "Recently, you have gone through some experiences that have triggered negative emotions, and you shared these with your friend/partner to seek comfort and understanding." They were then asked to report their own emotional negativity while recalling each experience they themselves experienced. Next, regulators were instructed to write down what they would say or do (regulation strategies) to improve their targets' emotions for each scenario within 25 min in the tone of talking directly to your partner. The draft of the regulation strategies they wrote will be handed by the experimenter to the target in another room. Afterward, the targets assessed the suitability of the regulation strategies and reported their emotional negativity after reading them under the instruction of

> Here's the draft of what your friend/partner want to say and do for you to help you feel better. Please read it carefully and report your emotional negativity regarding each scenario after reading the strategies. Additionally, assessed suitability of the regulation strategies provided by your friend/partner.

The four scenarios were presented in random order during the task.

Regulator's Emotion Perception Accuracy. In the second step of the IER task, regulators estimated the negative intensity of the emotions the targets were likely experiencing, using a 5-point Likert scale ranging from 1 (*not negative at all*) to 5 (*very negative*) based on the four scenarios provided. At the same time, targets rated their own emotional negativity on the same scale, imagining how they would have felt in those situations. The regulator's emotion perception accuracy was defined as the degree to which the regulator's rating matched the target's self-rating. This was calculated as the absolute difference between the regulator's inference and the target's rating for each scenario. The average accuracy score across the four scenarios was used for analysis (similar approaches can be found in Israelashvili et al., 2020; Zhou et al., 2017). While it is a little bit difficult to interpret this index, larger absolute differences indicate lower accuracy. The average absolute difference was reversed (average absolute difference multiplied by −1), such that a higher score reflects more accurate emotion perception.

Target-Perceived Strategy Suitableness. In the third step of the IER task, targets were asked to rate the suitability of the strategies on a 5-point Likert scale, ranging from 1 (*not suitable for me at all*) to 5 (*very suitable for me*), based on the four given scenarios. A higher average rating score indicated that the strategies were perceived as more suitable by the target.

Emotion Regulation Effectiveness. The goal of IER is to influence the emotional state of the target (Nozaki & Mikolajczak, 2020; Zaki & Williams, 2013). Thus, the difference in the target's negative emotions before and after regulation serves as an objective indicator of IER effectiveness. After reading the provided strategies, targets were asked to rate their emotional negativity toward each scenario on the same 5-point Likert scale, ranging from 1 (*not negative at all*) to 5 (*very negative*). Emotion regulation effectiveness was calculated as the difference between targets' negative emotion ratings before and after reading the regulator's strategy for the same scenario (i.e., the rating before regulation minus the rating after regulation). A larger rating difference indicates that the regulator's strategy was more effective in reducing the targets' negative emotions.

Perceived Social Distance Between the Regulator and the Target

The Inclusion of Other in the Self Scale (Aron et al., 1992) was adopted to measure the perceived social distance between each adolescent and their partner in the pair. The Other in the Self Scale is widely recognized for assessing relationship closeness (e.g., Helgeson & Van Vleet, 2019; Pietras & Briken, 2021; K. Zhang et al., 2021). Participants rated their perceived social distance on a scale of 1 to 7 using a set of seven diagrams, each consisting of two circles. One circle was labeled "self" and the other "your partner." The diagrams ranged from two circles that just touched (score of 1, indicating *far social distance*) to nearly overlapping circles (score of 7, indicating

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close social distance). Participants selected the diagram that best represented their relationship with their partner. The social distance index for each dyad was calculated as the average of the closeness ratings provided by the two adolescents.

Procedure

The procedural timeline was shown in Figure 3. Data for the present study were collected between September and November 2022. In line with the hypothesis, empathy was conceptualized as the antecedent variable. Therefore, empathy data were collected prior to the IER task. After registering and confirming participation, participants signed an online informed consent form, with parental approval, and completed an online questionnaire that included demographic information and the GEM empathy scale. For the analysis, only empathy scores from participants who were assigned the role of "regulator" in the IER task were included. One week later, participants came to our lab for the IER task. Some participated with same-sex friends in their class, while others were randomly paired with same-gender strangers from their grade. All participants who completed the experiment received a set of stationery (worth 40 Chinese Yuan) as a reward.

Analytic Strategy

All data were entered and analyzed in Statistical Package for the Social Sciences 28.0. First, a manipulation check was conducted using an independent-samples t test to determine whether the perceived social distance was significantly closer for friend dyads compared to stranger dyads. Second, all continuous variables were standardized. Third, descriptive statistics (mean, standard deviation) were calculated for all variables, followed by Pearson's correlations and partial correlation among the variables. Given the interrelationships among various components of empathy, partial correlation analysis was conducted to explore the associations between each specific empathy component and other variables, with the influence of the other two components being controlled. Next, the PROCESS macro Version 4.2 for Statistical Package for the Social Sciences (Model 4) was used to test the mediating effect of regulator's emotion perception accuracy and target-perceived strategy suitability (Hayes, 2022). Finally, the PROCESS macro Version 4.2 (Model 8) was applied to examine the moderating effect of social distance on the relationships between antecedent variables and mediating variables, as well as the antecedent variables and outcome variables. All variables included in Model 4 and Model 8 were standardized. Notably, to examine the unique contribution of a specific empathy component to IER effectiveness, the other two empathy components were controlled for as covariates in both Model 4 and Model 8, analyzed using the PROCESS macro Version 4.2.

Transparency and Openness

We have reported how we determined the sample size and measures used in the study. Anonymized data, scripts, and study materials have been made publicly available on the Open Science Framework and can be accessed at https://osf.io/udpq8/ (Wang, 2025). This study has not been preregistered.

Results

Manipulation Check

The perceived social distance between the regulator and the target was significantly closer in the friend dyads compared to the stranger dyads ($M_{\text{friends}} = 5.96$, $SD = 0.77$, vs. $M_{\text{stranger}} = 3.31$, $SD = 1.09$, $t = 28.70$, $p < .001$, Cohen's $d = 2.83$), indicating successful manipulation. Based on the effective manipulation of social distance, social distance was treated as a continuous variable in the subsequent analysis, enabling a more nuanced examination of its moderating effect on the relationship between the regulator's trait empathy and the regulation effectiveness.

Descriptive Analyses and Bivariate Analyses

Means, standard deviations, and correlations for all variables are presented in Table 1. After controlling for regulator's behavioral empathy and affective empathy, regulator's cognitive empathy was positively associated with emotion perception accuracy ($r = 0.18$, $p = .001$) and regulation effectiveness ($r = 0.18$, $p = .001$), as well as perceived social distance between the regulator and target ($r = 0.14$, $p = .006$). Emotion perception accuracy was positively associated with regulation effectiveness ($r = 0.25$, $p < .001$). There was no significant relationship between emotion perception accuracy and social distance ($r = -0.01$, $p = .836$). After controlling for regulator's cognitive empathy and affective empathy, regulator's behavioral empathy was positively associated with target-perceived strategy suitability ($r = 0.24$, $p < .001$) and regulation effectiveness ($r = 0.27$, $p < .001$). There

Figure 3
The Procedure of the Present Study

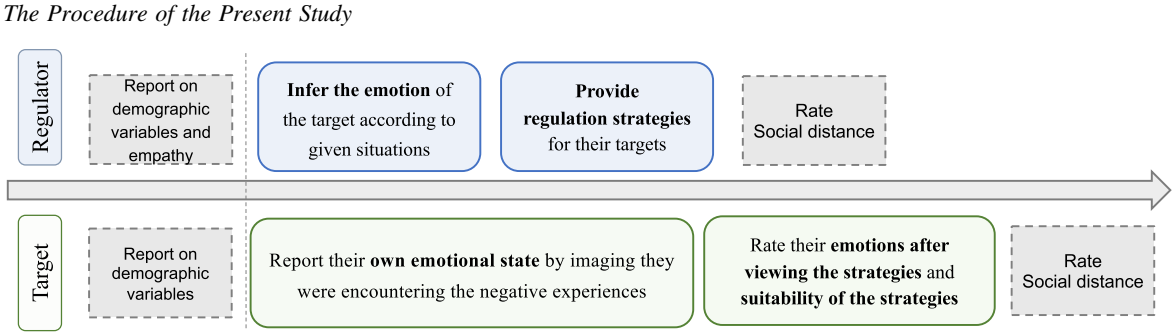

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Table 1
Descriptive Statistics and Correlations Among Variables

Variable	*M*	*SD*	1	2	3	4	5	6	7
1. Regulator's cognitive empathy	1.07	1.21	—						
2. Regulator's behavioral empathy	0.99	1.71	−.06	—					
3. Regulator's affective empathy	0.38	1.99	.17*	−.03	—				
4. Regulator's emotion perception accuracy	−1.30	0.74	.18**	−.02	.04	—			
5. Target-perceived strategy suitability	3.55	1.16	.11*	.24***	.18**	−0.09	—		
6. Regulation effectiveness	0.95	0.91	.18**	.27***	0.01	0.25***	0.43***	—	
7. Social distance	4.65	1.62	.14**	.05	0.34***	−0.01	0.46***	0.29***	—

Note. The relationships presented in the table between a specific component of empathy and other variables are partial correlations, controlling for the other two components of empathy.
* $p < .05$. ** $p < .01$. *** $p < .001$.

was no significant correlation between regulator's behavioral empathy and social distance ($r = 0.05$, $p = .346$) after controlling for the other two empathy components. Target-perceived strategy suitability was positively associated with regulation effectiveness ($r = 0.43$, $p < .001$) and social distance ($r = 0.46$, $p < .001$). Regulation effectiveness was positively associated with social distance ($r = 0.29$, $p < .001$). In addition, regulator's affective empathy was not significantly associated with regulation effectiveness after controlling for cognitive empathy and behavioral empathy ($r = 0.01$, $p = .878$).

Testing for the Mediation Effect

In Hypothesis 1, this study proposed that regulator's emotion perception accuracy would mediate the relationship between regulator's cognitive empathy and regulation effectiveness. Model 4 of the PROCESS macro Version 4.2 (Hayes, 2022) was used to examine this hypothesis, which involves three steps: (a) assessing whether there is a significant link between regulator's cognitive empathy and emotion perception accuracy, (b) examining whether there is a significant link between regulator's emotion perception accuracy and regulation effectiveness, and (c) assessing whether the relationship between the regulator's cognitive empathy and regulation effectiveness remains significant when controlling for emotion perception accuracy. As shown in Table 2 and Figure 4, after controlling for regulator's behavioral empathy and affective empathy, cognitive empathy was found to be positively associated with emotion perception accuracy ($b = 0.18$, $p = .001$), which in turn was positively related to regulation effectiveness ($b = 0.27$, $p < .001$). Simultaneously, the residual direct effect of regulator's cognitive empathy on regulation effectiveness ($b = 0.12$, $p = .018$) was also significant. Therefore, regulator's emotion recognition accuracy mediated the relationship between cognitive empathy and regulation effectiveness (indirect effect = 0.05, 95% CI [0.02, 0.08]; more detailed information was presented in Supplemental Table S2). The mediation effect accounted for 28% of the total effect of cognitive empathy on regulation effectiveness. Thus, Hypothesis 1 was supported.

Similarly, Model 4 of the PROCESS macro Version 4.2 (Hayes, 2022) was utilized to test Hypothesis 2 of the present study, which proposed that target-perceived strategy suitability mediates the relationship between the regulator's behavioral empathy and regulation effectiveness. As shown in Table 3 and Figure 5, after controlling for regulator's cognitive empathy and affective empathy, the result showed that regulator's behavioral empathy significantly positively predicted target-perceived strategy suitability ($b = 0.23$, $p < .001$) and target-perceived strategy suitability was significantly associated with regulation effectiveness ($b = 0.33$, $p < .001$). The positive direct association between regulator's behavioral empathy and regulation effectiveness remained still significant ($b = 0.20$, $p = .001$). The indirect effect of regulator's behavioral empathy on regulation effectiveness was mediated by target-perceived strategy suitability (indirect effect = 0.08, 95% CI [0.04, 0.12]; more detailed information was presented in Supplemental Table S3). The mediation effect explained 28% of the total effect of regulator's behavioral empathy on regulation effectiveness. Therefore, Hypothesis 2 was supported.

Table 2
Testing the Mediating Effect of Emotion Perception Accuracy in the Relationship Between Regulator's Cognitive Empathy and Regulation Effectiveness

	Regulator's emotion perception accuracy		Interpersonal emotion regulation effectiveness	
Predictor	*b*	*t*	*b*	*t*
Regulator's behavioral empathy	0.01	0.02	0.26	5.95***
Regulator's affective empathy	0.03	0.44	0.01	0.02
Regulator's cognitive empathy	0.18	3.24**	0.12	2.38*
Regulator's emotion perception accuracy			0.27	5.95***
R^2	0.04		0.17	
F	4.11**		13.46***	

* $p < .05$. ** $p < .01$. *** $p < .001$.

Testing for the Moderated Mediation

As noted, Hypothesis 3 proposed that social distance would moderate the associations between regulator's cognitive empathy and emotion perception accuracy as well as regulator's cognitive empathy and regulation effectiveness. To examine this hypothesis, we estimated parameters for two regression models with Hayes's (2022) PROCESS macro Version 4.2 (Model 8). Specifically, we estimated the moderating effect of social distance on (a) the relationship between regulator's cognitive empathy and emotion perception accuracy and (b) the relationship between regulator's cognitive empathy and regulation effectiveness. As Table 4 illustrates, after

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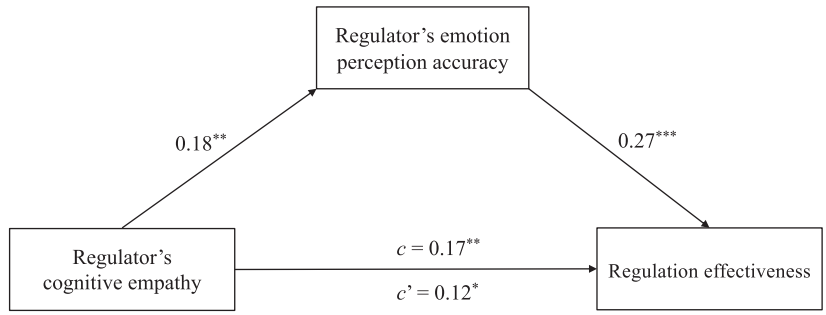
Figure 4

The Mediation Model of Regulator's Emotion Perception Accuracy Between Regulator's Cognitive Empathy and Regulation Effectiveness



Note. The coefficients of each path were standardized. *c* means the total effect between cognitive empathy and regulation effectiveness; *c′* means the indirect effect.
$^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

controlling for regulator's behavioral empathy and affective empathy, social distance moderated the relationship between regulator's cognitive empathy and emotion perception accuracy ($b = 0.14$, $p = .012$). For descriptive purposes, we plotted regulator's emotion perception accuracy against regulator's cognitive empathy, separately for close and far social distance (Figure 6A). When social distance was close ($M + 1$ *SD*), regulator's cognitive empathy significantly positively predicted emotion perception accuracy ($b = 0.33$, $p < .001$). However, this association was not significant when social distance was far ($M - 1$ *SD*; $b = 0.05$, $p = .567$). The detailed result of the path coefficient was shown in Table 5. Furthermore, Figure 6B depicted the conditional relationship between the regulator's cognitive empathy and emotion perception accuracy across varying levels of social distance, demonstrating that as the social distance between the regulator and the target becomes closer, the positive effect of regulator's cognitive empathy on emotion perception accuracy became more pronounced. Specifically, reduced social distance allows the regulator's cognitive empathy to better enhance the accuracy of emotion perception. In addition, we also tested whether social distance moderated the relationship between regulator's cognitive empathy and regulation effectiveness as proposed in Hypothesis 3. However, the moderating role of social distance on the relationship between regulator's cognitive empathy and regulation effectiveness was not significant ($b = -0.07$, $p = .183$; Tables 4 and 5). Therefore, Hypothesis 3 was partly supported.

When the initial segment of a mediation pathway is influenced by a moderator variable, it is worthwhile to further explore whether the indirect effect of emotion perception accuracy on the relationship between the regulator's cognitive empathy and regulation effectiveness was significant in cases where social distance (the moderator variable) was close and far. Supplemental Table S4 presents the result, demonstrating that the indirect effect of emotion perception accuracy on the relationship between the regulator's cognitive empathy and regulation effectiveness was significant at close social distance ($M + 1$ *SD*, indirect effect = 0.10, 95% CI [0.05, 0.16]). However, the mediation effect was not significant at far social distance ($M - 1$ *SD*, indirect effect = 0.02, 95% CI [−0.04, 0.07]).

Moreover, Hypothesis 4 proposed that social distance would moderate the associations between regulator's behavioral empathy and target-perceived strategy suitability as well as regulator's behavioral empathy and regulation effectiveness. Similarly, Model 8 of the PROCESS macro Version 4.2 (Hayes, 2022) was utilized to test this hypothesis. Specifically, we estimated the moderating effect of social distance on (a) the relationship between regulator's behavioral empathy and target-perceived strategy suitability and (b) the relationship between regulator's behavioral empathy and regulation effectiveness (Table 6). Consistent with Hypothesis 4, after controlling for regulator's cognitive empathy and affective empathy, social distance moderated the relationship between regulator's behavioral empathy and regulation effectiveness ($b = 0.23$, $p < .001$). For the descriptive purpose, we plotted regulation effectiveness against regulator's behavioral empathy, separately for close and far social distance (Figure 7A). Specifically, when social distance was close ($M + 1$ *SD*), regulator's behavioral empathy significantly positively predicted regulation effectiveness ($b = 0.34$, $p < .001$). However, when social distance became far ($M - 1$ *SD*), this association was not significant ($b = -0.03$, $p = .752$). The detailed result of the path coefficient was shown in Table 7. Figure 7a depicted the conditional relationship between the regulator's behavioral empathy and regulation effectiveness across varying levels of social distance, showing that as the social distance between the regulator and the target became closer, the effect

Table 3

Testing the Mediating Effect of Target-Perceived Strategy Suitability in the Relationship Between Regulator's Behavioral Empathy and Regulation Effectiveness

Predictor	Target-perceived strategy suitability		Regulation effectiveness	
	b	*t*	*b*	*t*
Regulator's cognitive empathy	0.09	1.49	0.14	2.52*
Regulator's affective empathy	0.16	3.01**	−0.04	−0.73
Regulator's behavioral empathy	0.23	4.48**	0.20	3.28**
Target-perceived strategy suitability			0.33	6.01***
R^2	0.09		0.20	
F	11.91***		14.76***	

$^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

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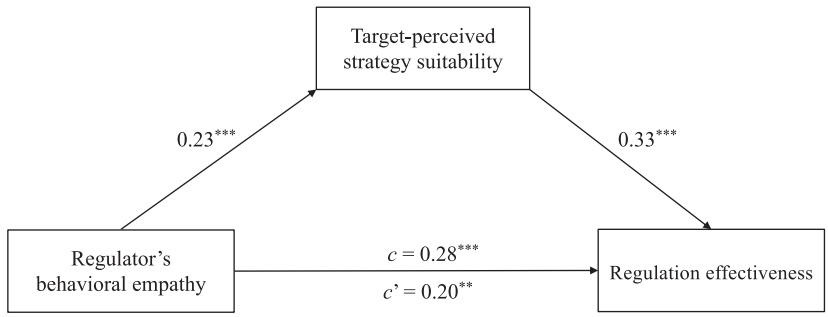
Figure 5

The Mediation Model of Target-Perceived Strategy Suitability Between Regulator's Behavioral Empathy and Regulation Effectiveness



Note. The coefficients of each path were standardized. *c* means the total effect between cognitive empathy and regulation effectiveness; *c′* means the indirect effect.
** $p < .01$. *** $p < .001$.

of regulator's behavioral empathy on regulation effectiveness became more pronounced.

In addition, as predicted in Hypothesis 4, social distance also moderated the relationship between regulator's behavioral empathy and target-perceived strategy suitability ($b = -0.11$, $p = .006$). However, the moderating way of social distance was not consistent with the proposed hypothesis (Figure 7B). Specifically, when social distance was far ($M - 1$ *SD*), regulator's behavioral empathy significantly positively predicted target-perceived strategy suitability ($b = 0.33$, $p < .001$). When social distance became close ($M + 1$ *SD*), this positive association was also significant, but much weaker ($b = 0.13$, $p = .012$). The detailed result of the path coefficient was shown in Table 7. Figure 7b depicted the conditional relationship between the regulator's behavioral empathy and target-perceived strategy suitability across varying levels of social distance, demonstrating that as the social distance between the regulator and the target became far, the positive effect of regulator's behavioral empathy on target-perceived strategy suitability became more pronounced. Thus, Hypothesis 4 was partly supported.

Table 4

Testing the Moderating Effect of Social Distance on the Relationship Between Regulator's Cognitive Empathy and Regulation Effectiveness via Emotion Perception Accuracy

Predictor	Regulator's emotion perception accuracy		Regulation effectiveness	
	b	*t*	*b*	*t*
Regulator's behavioral empathy	0.02	0.32	0.24	3.99***
Regulator's affective empathy	0.05	0.87	−0.08	−1.62
Regulator's cognitive empathy	0.19	3.46**	0.08	1.55
Social distance	−0.12	−2.00*	0.28	4.95***
Regulator's Cognitive Empathy × Social Distance	0.14	2.03*	−0.07	−1.33
Regulator's emotion perception accuracy			0.30	6.74***
R^2	0.24**		0.23***	
F	4.43		17.40	

* $p < .05$. ** $p < .01$. *** $p < .001$.

We have further explored whether the indirect effect of target-perceived strategy suitability on the relationship between the regulator's behavioral empathy and regulation effectiveness was significant in cases where social distance (the moderator variable) was close and far. Supplemental Table S5 presented the result, demonstrating that the indirect effect of target-perceived strategy suitability on the relationship between the regulator's behavioral empathy and regulation effectiveness was significant at far social distance ($M - 1$ *SD*, indirect effect = 0.10, 95% CI [0.06, 0.16]). While the mediation effect remained significant when social distance was close ($M + 1$ *SD*), its magnitude was substantially weaker (indirect effect = 0.04, 95% CI [0.01, 0.07]).

In the preceding analyses, we consistently employed the perceived social distance between the regulator and the target as the moderating variable. With regard to the experimental design, we implemented a binary manipulation of the relationship between the regulator and the target, categorizing them as either friend dyads or stranger dyads. Thus, we also treated social distance as a binary variable and examined the mediating role of emotion perception accuracy and target-perceived strategy suitability separately for friend dyads and stranger dyads. The results were consistent with the preceding analysis, further confirming the robustness of the current findings. Detailed results are provided in the Supplemental Materials.

Discussion

Emotional health plays a critical role in adolescent development. Given the heightened sensitivity to peers' emotions and behaviors during adolescence, IER by peers may complement self-regulation and enhance emotion regulation effectiveness (Sahi et al., 2023). Previous research has consistently highlighted the role of trait empathy in improving IER (Hallam et al., 2014; Morawetz et al., 2021; Reeck et al., 2016; Zaki, 2020). However, the specific contributions of different empathy components and underlying mechanisms remain underexplored. Building on the extended process model of extrinsic emotion regulation and social baseline theory, the present study tested a moderated mediation model to examine how regulator's trait empathy, particularly cognitive and behavioral empathy, influences emotion regulation effectiveness through emotion perception accuracy and

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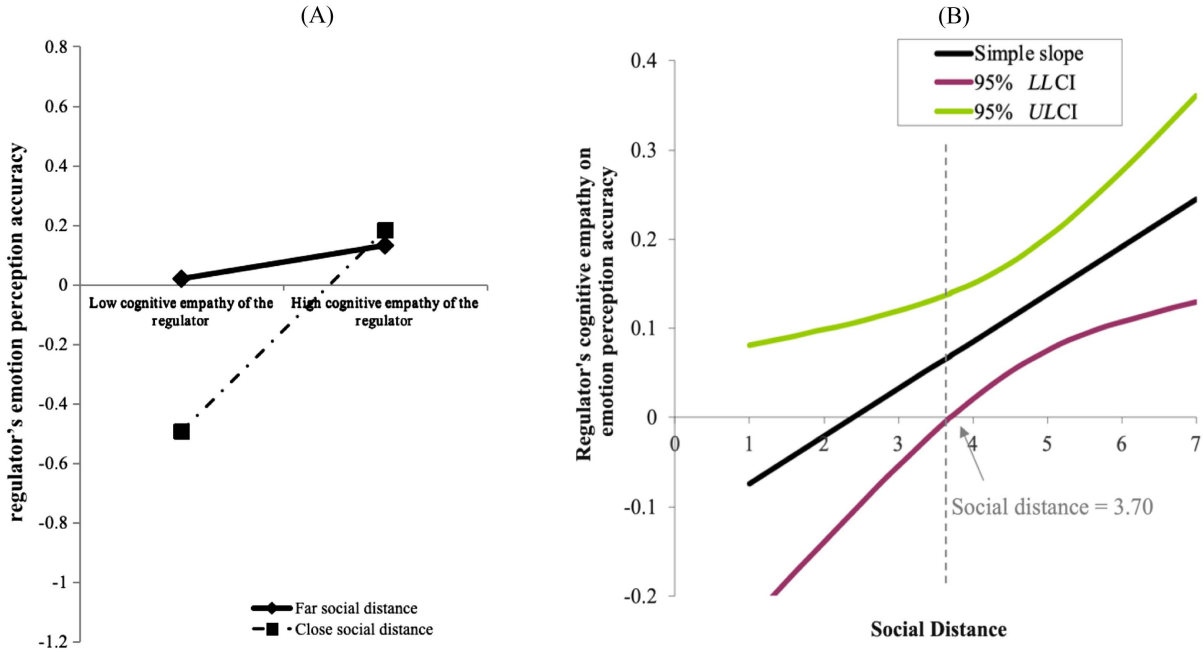
Figure 6

Regulator's Emotion Perception Accuracy as a Function of Regulator's Cognitive Empathy and Social Distance



Note. (A) Functions are graphed for two levels of social distance: "close social distance" means 1 standard deviation above the mean, and "far social distance" means 1 standard deviation below the mean. Note that the variables in the analysis are standardized, and the graph is for descriptive purposes only. (B) The *y*-axis indicates the slope for the effect of regulator's cognitive empathy on emotion perception accuracy for each point of social distance. The dark black line is the simple regression line showing the effect of regulator's cognitive empathy on emotion perception accuracy by social distance, and the two curves are the 95% confidence bands. Areas to the right of the dotted line represent the regions of significance for the conditional association. CI = confidence interval; *LL* = lower limit; *UL* = upper limit. See the online article for the color version of this figure.

target-perceived strategy suitability, respectively, and how social distance moderates these relationships. The results have shown that cognitive empathy positively predicted regulation effectiveness, with emotion perception accuracy partially mediating this effect. Similarly, behavioral empathy positively predicted regulation effectiveness, with target-perceived strategy suitability partially mediating the relationship. Additionally, social distance moderated the links between cognitive empathy and emotion perception accuracy, behavioral empathy and target-perceived strategy suitability, and behavioral empathy and regulation effectiveness.

Table 5

Coefficient of Each Path in the Relationship Between Cognitive Empathy and Regulation Effectiveness

Path	Social distance	*b*	*SE*	95% CI
Regulator's cognitive empathy → Emotion perception accuracy	$M - 1\ SD$	0.05	0.09	[−0.13, 0.24]
	M	0.19	0.06	[0.08, 0.30]
	$M + 1\ SD$	0.33	0.08	[0.17, 0.49]
Regulator's cognitive empathy → Regulation effectiveness	$M - 1\ SD$	0.15	0.07	[0.01, 0.29]
	M	0.08	0.05	[−0.02, 0.18]
	$M + 1\ SD$	0.01	0.07	[−0.12, 0.15]

Note. The relationship between regulator's cognitive empathy and emotion perception accuracy as well as the relationship between regulator's cognitive empathy and regulation effectiveness on different levels of social distance. $M + 1\ SD$ = close social distance; $M - 1\ SD$ = far social distance; *SE* = standard error; CI = confidence interval.

Regulator's Empathy and Regulation Effectiveness

As anticipated, both cognitive and behavioral empathy facilitate regulation effectiveness. This highlights the importance of the regulator's social cognition in shaping effective IER. These findings align with previous research showing that regulators with higher other-oriented empathy are more likely to help targets improve their negative emotions (Chavira Trujillo et al., 2022). Specifically, regulators who better understand and care for their targets' emotions tend to be more effective in improving their targets' emotional states. This result also supports Reeck et al.'s (2016) hypothesis that brain regions related to empathy may play a role in IER. Additionally, these findings can be understood in the context of empathy development during adolescence. As cognitive and behavioral empathy are still developing in adolescence (Wang et al., 2021), individual differences in the two components often emerge in social interactions.

While affective empathy is prominent during adolescence, it does not relate to regulation effectiveness. This may be because regulators with high affective empathy are more likely to share the target's negative emotions, leading to self-oriented distress that interferes with effective regulation (Weisz & Cikara, 2021; Zaki, 2020). Consistent with this, research has shown that regulators with high affective empathy experience stronger sympathetic nervous system reactivity and greater self-oriented negative affect in response to a target's distress (Brown et al., 2020).

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Table 6

Testing the Moderating Effect of Social Distance on the Relationship Between Regulator's Behavioral Empathy and Regulation Effectiveness via Target-Perceived Strategy Suitability

Predictor	Target-perceived strategy suitability		Regulation effectiveness	
	b	*t*	*b*	*t*
Regulator's cognitive empathy	0.03	0.57	0.12	2.27*
Regulator's affective empathy	0.02	0.35	−0.05	−1.06
Regulator's behavioral empathy	0.23	5.34***	0.16	2.64**
Social distance	0.42	8.39***	0.14	2.43*
Regulator's Behavioral Empathy × Social Distance	−0.11	−2.77**	0.19	3.14**
Target-perceived strategy suitability			0.31	5.29***
R^2	0.26***		0.25***	
F	29.99		12.09	

* $p < .05$. ** $p < .01$. *** $p < .001$.

The Mediating Effect of Emotion Perception Accuracy in the Relationship Between Regulator's Cognitive Empathy and Regulation Effectiveness

This study is the first to identify emotion perception accuracy as a mediator in the link between the regulator's cognitive empathy and IER effectiveness. Specifically, cognitive empathy enhances emotion perception accuracy, which in turn improves regulation effectiveness. Thus, emotion perception accuracy serves as both an outcome of cognitive empathy and a driver of regulation success. However, emotion perception accuracy only partially mediates this relationship, suggesting that cognitive empathy may also directly influence regulation effectiveness.

In addition to the overall mediation result, each individual path in the model warrants attention. For the first path, the findings support the hypothesis that the regulator's cognitive empathy is positively related to emotion perception accuracy. Specifically, accurate perception of the target's emotions depends on the regulator's ability to understand emotions through top-down processes. This aligns with Reeck et al.'s (2016) theoretical model, which suggests that neural systems involved in cognitive empathy also play a role in emotion recognition and IER. Furthermore, an functional magnetic resonance imaging study with adolescents supports this by showing that greater activation in regions associated with perspective-taking, such as the medial prefrontal cortex and temporoparietal junction, is linked to better emotion perception performance (Kral et al., 2017). A meta-analysis of nearly sixty studies also found a significant positive relationship between cognitive empathy and emotion recognition accuracy (Qiao et al., 2025).

For the second path in the mediation model, the present study has found that the regulator's emotion perception accuracy was positively associated with regulation effectiveness. According to the extended process model of extrinsic emotion regulation, accurately perceiving the target's emotions is the first step in effective regulation (Nozaki & Mikolajczak, 2020). Consistent with this theory, prior studies have shown that accurate emotion perception enhances relationship satisfaction in adolescents' social interactions, whether with romantic partners or peers (Berlamont et al., 2023; Haugen et al., 2008). The current findings have, to some extent, explained the findings of the mentioned studies. Specifically, accurate emotion perception leads to effective regulation, which in turn improves the targets' relationship satisfaction.

The Mediating Effect of Target-Perceived Strategy Suitability in the Relationship Between Regulator's Behavioral Empathy and Regulation Effectiveness

The study has also found that target-perceived strategy suitability mediated the relationship between the regulator's behavioral empathy and regulation effectiveness. Specifically, regulators who show greater empathy for the target's suffering and have a helping motivation are more likely to offer suitable strategies, which in turn enhances regulation effectiveness. Thus, target-perceived strategy suitability serves as a behavioral mechanism linking empathy to regulation outcomes. Notably, target-perceived strategy suitability only partially mediates this relationship, suggesting that cognitive empathy may also directly predict regulation effectiveness.

The first path of the mediation supports the hypothesis that the regulator's behavioral empathy is positively associated with target-perceived strategy suitability. Specifically, regulators' tendency to express the understanding of target's negative experience through explicit behavior is essential for providing suitable regulation strategies. This is likely because regulators with higher behavioral empathy are more motivated to help improve the target's emotional state, and offering appropriate strategies that meet the target's needs is the most direct way to demonstrate this motivation. Although previous research has not directly examined the role of behavioral empathy in regulation effectiveness, it has shown that the tendency of empathic concern promotes the use of appropriate emotion regulation strategies and facilitates helping behaviors (Chavira Trujillo et al., 2022; Profe et al., 2021; Van der Graaff et al., 2018). Empathic concern, though typically categorized as the other-focused affective empathy (Chavira Trujillo et al., 2022), embodies the drive to care for others' negative experiences and frequently translates into prosocial behavioral responses (Batson, 1991; Batson et al., 1987; de Waal, 2008). Thus, findings on the effect of empathic concern on IER provide preliminary support for the relationship between behavioral empathy and strategy suitability. Since empathic concern, as emphasized in previous studies, is not equivalent to behavioral empathy, the findings derived from the more direct assessment of behavioral empathy in the present study warrant greater attention.

The second path of the mediation model reveals that target-perceived strategy suitability is positively associated with regulation effectiveness. According to the extended process model of extrinsic emotion regulation, selecting and implementing appropriate strategies is crucial for effective regulation (Nozaki & Mikolajczak, 2020). Supporting this, previous studies have shown that adaptive emotion regulation strategies from peers reduce negative emotions in adolescents (Do et al., 2025). Additionally, peer social support, a form of responsive and adaptive regulation, has demonstrated both immediate and long-term benefits in alleviating adolescent psychological distress (Lyell et al., 2020). This study further confirms that target-perceived strategy suitability serves as a bridge between the regulator's behavioral empathy and regulation effectiveness.

The Moderating Effect of Social Distance

As hypothesized, social distance moderates the relationship between cognitive empathy and emotion perception accuracy. Specifically, for

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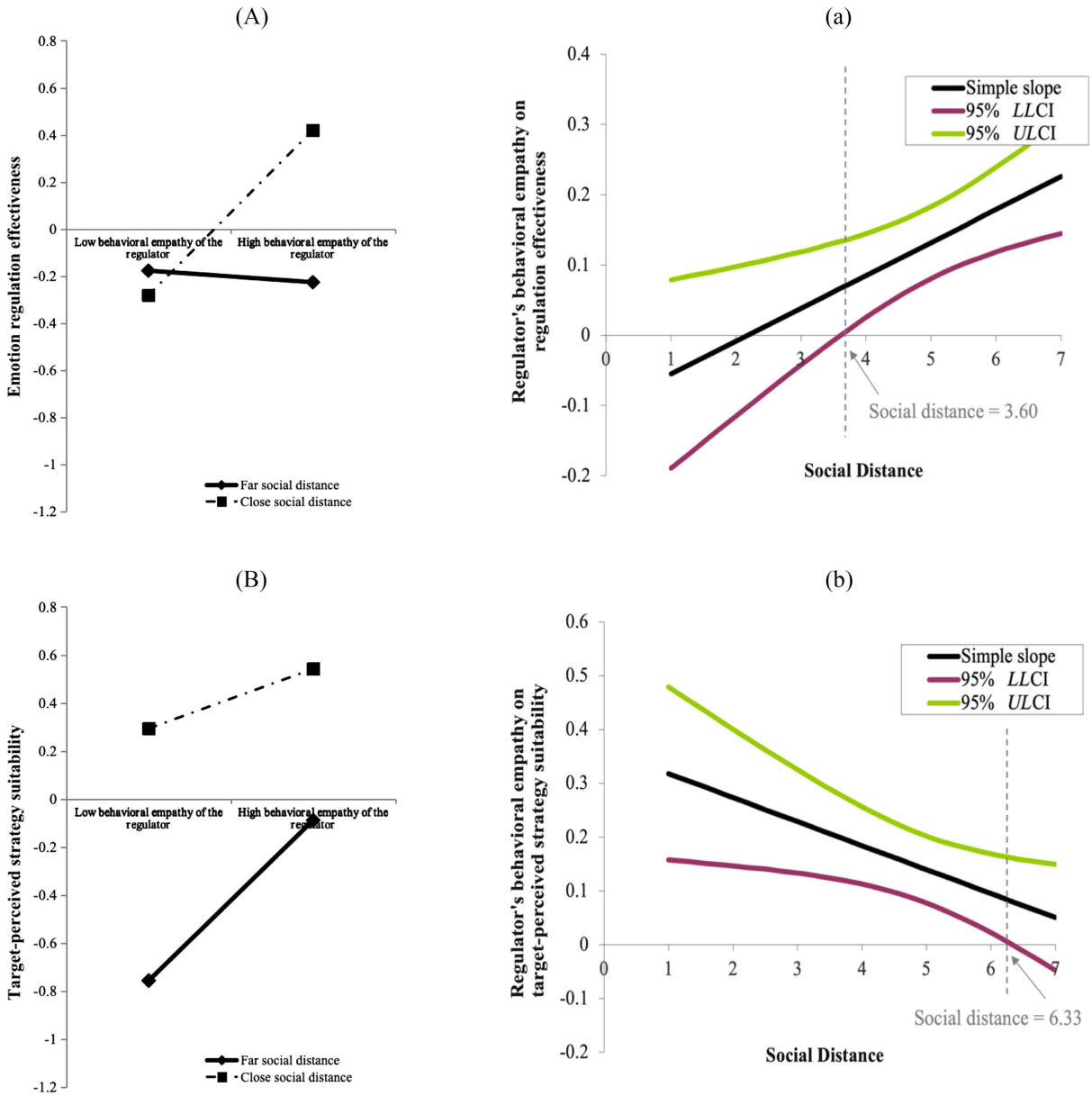
Figure 7

Target-Perceived Strategy Suitability as a Function of Regulator's Behavioral Empathy and Social Distance



Note. Emotion regulation effectiveness as a function of regulator's behavioral empathy and social distance. (A, B) Functions are graphed for two levels of social distance: "close social distance" means 1 standard deviation above the mean, and "far social distance" means 1 standard deviation below the mean. Note that the variables in the analysis are standardized, and the graph is for descriptive purposes only. (a) The *y*-axis indicates the slope for the effect of regulator's behavioral empathy on target-perceived strategy suitability (b: regulator's behavioral empathy on regulation effectiveness) for each point of social distance. The dark black line is the simple regression line showing the effect of regulator's behavioral empathy on target-perceived strategy suitability (b: regulator's behavioral empathy on regulation effectiveness) by social distance, and the two curves are the 95% confidence bands. Areas to the left of a (the right of b) on the dotted line represent the regions of significance for the conditional association. CI = confidence interval; *LL* = lower limit; *UL* = upper limit. See the online article for the color version of this figure.

adolescent dyads with close social distance, cognitive empathy is significantly associated with more accurate emotion perception. However, for those with far social distance, the relationship is not significant. It suggests that closer social distance enhances the positive effect of cognitive empathy on accurately perceiving the target's negative emotions. This can be explained by familiarity, which boosts empathy motivation. According to social baseline theory, familiar targets' emotions are easier to process and understand, saving cognitive resources (Beckes & Sbarra, 2022; Cameron et al., 2019). In line with this, research has shown that empathy-driven helping behavior is more

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Table 7
Coefficient of Each Path in the Relationship Between Behavioral Empathy and Regulation Effectiveness

Path	Social distance	*b*	*SE*	95% CI
Regulator's behavioral empathy → Regulation effectiveness	$M - 1\ SD$	−0.03	0.09	[−0.20, 0.15]
	M	0.16	0.06	[0.04, 0.27]
	$M + 1\ SD$	0.34	0.08	[0.19, 0.49]
Regulator's behavioral empathy → Target-perceived strategy suitability	$M - 1\ SD$	0.33	0.06	[0.21, 0.46]
	M	0.23	0.04	[0.14, 0.31]
	$M + 1\ SD$	0.13	0.05	[0.03, 0.22]

Note. The relationship between regulator's behavioral empathy and regulation effectiveness as well as the relationship between regulator's behavioral empathy and target-perceived strategy suitability on different levels of social distance. $M + 1\ SD$ = close social distance; $M - 1\ SD$ = far social distance; *SE* = standard error; CI = confidence interval.

pronounced when the target is familiar (Siem, 2022). Moreover, adolescents are sensitive to the costs and benefits of social interactions (Padilla-Walker & Fraser, 2014). Therefore, close social distance may act as a booster to enhance the positive function of regulator's cognitive empathy on emotion perception accuracy.

However, contrary to the proposed hypothesis, social distance does not moderate the relationship between cognitive empathy and regulation effectiveness. It also means that whether the target is a friend or a stranger, cognitive empathy remains crucial for effective emotion regulation. Previous studies, including an functional magnetic resonance imaging study, suggest that brain regions associated with cognitive empathy are activated during IER, even with strangers (Hallam et al., 2014). The present study provides further evidence that the promoting effect of social distance on the relationship between regulator's cognitive empathy and regulation effectiveness is primarily reflected in enhancing the association between regulator's cognitive empathy and emotion perception accuracy among familiar friends.

In line with the proposed hypothesis, social distance moderates the effect of the regulator's behavioral empathy on the regulation effectiveness. It has been shown that the direct association between behavioral empathy and regulation effectiveness is only significant when the social distance between the regulator and the target is close. In contrast, no significant relationship exists between behavioral empathy and regulation effectiveness when the social distance is far. In other words, closer social distance strengthens the positive effect of behavioral empathy on regulatory outcomes. This can be explained by the social baseline theory, which posits that regulators invest more empathy in familiar targets (Beckes & Coan, 2011; Beckes & Sbarra, 2022). Consistent findings suggest that regulators demonstrate more care when regulating the emotions of close targets (Tanna & MacCann, 2023). In addition, behavioral empathy, which often manifests through explicit empathic responses, is more readily perceived by those who are close rather than by those who are distant (Wang et al., 2021). In familiar relationships, targets are more likely to draw on past interactions to recognize the peer regulators' empathic tendencies to attend to or comfort them, which is sufficient to alleviate their negative emotions.

Contrary to the proposed hypothesis, the moderating effect of social distance on the relationship between behavioral empathy and the target-perceived strategy suitability shows a different pattern. Specifically, when social distance is far, behavioral empathy has a significant impact on target-perceived strategy suitability. While the relationship between behavioral empathy and perceived strategy suitability is much weaker when social distance is close. It can be explained that due to the limited interaction experiences with the unfamiliar regulator, the target cannot directly benefit from the regulator's behavioral empathy. Whether appropriate strategies are provided becomes a crucial factor for the targets to evaluate the IER effectiveness. Related studies have shown that adolescents may display caring tendencies toward strangers, which in turn relate to their prosocial behavior to improve stranger's well-being (Berry et al., 2018; Streit et al., 2020; Yaghoubi Jami et al., 2023). However, when social distance is close, the target's assessment of IER effectiveness often stems directly from the empathy traits of the familiar regulator, thereby reducing the relative importance of strategy appropriateness.

Theoretical and Practical Implications

Regarding the theoretical perspective, first, although the extended theoretical model of extrinsic emotion regulation suggests that IER effectiveness may be shaped by mediating factors such as emotion perception accuracy and strategy suitability, as well as individual differences in regulators' social cognitive traits, there remains a notable lack of empirical research to thoroughly validate these hypotheses with supporting data. This study incorporates the regulator's empathic traits into the extended model of extrinsic emotion regulation and reveals that different components of the regulator's trait empathy impact regulation outcomes through distinct processes, thereby deepening our theoretical understanding. Second, this study shifts the focus from the previously predominant emphasis on the harmful or maladaptive effects of peer influence (e.g., an increase in risky decision making and substance use in the presence of peers, Chein et al., 2011; Doremus-Fitzwater et al., 2010) to examining the positive impact of peers on adolescent emotional adaptation and uncovering the mechanisms through which such influence occurs.

Regarding practical implications, the findings offer valuable insights for interventions aimed at improving adolescents' emotional regulation effectiveness. Empathy, as a key antecedent influencing the IER effectiveness, serves as a valuable target spot for intervention. Mediating factors identified in the study, including emotion perception accuracy and strategy suitability, can be treated as the foundation for monitoring intervention outcomes. Furthermore, the study examines the boundary conditions, specifically social distance, under which the regulator's trait empathy impacts regulation effectiveness. It helps to identify specific social contexts in which focusing on particular components of empathy could more effectively improve regulation outcomes. Finally, the dyadic paradigm used in the present study can serve as a model for intervention activities, where dynamic interactions between regulators and targets foster social cognitive skills and enhance regulation effectiveness.

Limitations and Future Directions

The present study has several limitations that highlight potential avenues for future research. First, the IER task could be further refined. Specifically, the emotion-provoking events utilized in the task were not experienced by the targets in real time, which may create challenges for them when assessing their emotional states both during

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the events and after being regulated. While using predetermined events does help minimize the influence of extraneous variables, such as the content and the ways in which targets describe their negative experiences, this approach limits the immediacy of the emotional response. To address this limitation, we select events that are commonly encountered by adolescents, based on a prior coding study, and we carefully design the task instructions to enhance targets' immersion in the experience. However, the ecological validity could still be improved. Future research might enhance this by using daily diary methods to explore peer emotion regulation in naturalistic settings (Gadassi-Polack et al., 2025) and allowing participants to recall personal experiences based on specific prompts. Additionally, incorporating objective measures like heart rate variability assessed before and after emotion regulation could provide physiological indicators of emotional changes, offering a more precise evaluation of regulatory effectiveness. Another limitation of the task design is that we have assessed the effectiveness of regulation based on the changes of targets' overall emotional negativity rather than focusing on specific negative emotions, such as sadness, anger, or fear. Future research could investigate how regulators' empathy influences the regulation of these specific types of negative emotions, thereby enriching our understanding of the nuances in IER.

Second, the GEM employed in the present study is not a widely used scale for measuring adolescent trait empathy in previous research. While prior studies have largely focused on the cognitive and affective components of trait empathy in regulators, the behavioral component, representing the explicit manifestation of empathy that facilitates effective IER, has been overlooked. In this study, the adoption of the three-dimensional GEM scale is primarily aimed at exploring the role of behavioral empathy in IER, in addition to cognitive and affective empathy. Although the three-factor model shows slightly superior performance on certain fit indices, such as a lower root-mean-square residual and a higher comparative fit index, this version of the GEM has not been widely utilized in previous studies. Future research should consider utilizing well-established scales, such as the Interpersonal Reactivity Index (Davis, 1983) and the Basic Empathy Scale (Jolliffe & Farrington, 2006), to further verify the effects of cognitive and affective empathy. Additionally, to strengthen the findings related to behavioral empathy, it is essential to integrate parent- or peer-reported GEM data, or use observational indicators, to further validate the robustness of the results.

Third, given the cross-sectional nature of this study, the causal relationships among variables should be interpreted with caution. While a dyadic design is employed to collect data from both regulators and targets, addressing some limitations of cross-sectional research, future studies could adopt longitudinal designs to better understand the effects of peer regulators' social cognitive ability on the IER and emotional well-being of the adolescent target over time.

Fourth, the generalizability of the findings warrants further attention. The present study is based solely on an adolescent sample from one city, which limits the applicability of the results. Additionally, this study is not preregistered. While we have a relatively solid theoretical foundation, the moderating effect of social distance on the relationship between the regulator's behavioral empathy and regulation effectiveness does not fully align with our initial hypothesis. Moving forward, we will prioritize preregistration for future research. We also encourage subsequent studies to explore the moderating role of social distance by manipulating it across various levels (e.g., close friends, acquaintances, strangers) and employing a more diverse range of samples to ensure the robustness of the effect (Beckes & Sbarra, 2022).

Conclusions

Using a dyadic design, the present study investigates the role of regulator's trait empathy on IER effectiveness among adolescents and related behavioral mechanisms. The current findings have shown that regulators with higher cognitive empathy tend to perceive targets' negative emotions more accurately, which enhances IER effectiveness. Similarly, regulators with higher behavioral empathy offer more suitable strategies, boosting regulation success. Moreover, close social distance strengthens the positive effects of both cognitive empathy on emotion perception accuracy and behavioral empathy on regulation effectiveness, while far social distance amplifies the positive impact of behavioral empathy on target-perceived strategy suitability. This study deepens our understanding of the extended process model of extrinsic emotion regulation and emphasizes the importance of adolescent social cognition in facilitating IER success. Future research could further explore how regulators' trait empathy influences the regulation of specific types of negative emotions by daily diary approaches.

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